ZOVAK · PRODUCT PLAN

Haulix — the AI operations desk for logistics

What can be automated in a logistics workload, how the agents work, what stays with people, and how we build the backend on what we learned from Nofa / ZOVAK.

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30 Oct 2026 · Frontend demo built · backend not started

01 Summary

Haulix is an AI operations desk for freight forwarders, 3PLs, courier and trucking companies. Nine specialist agents take orders in any format, quote lanes, plan routes, dispatch drivers, run the warehouse, watch every shipment, fix problems, prepare customs documents and close out billing.

The rule is the same one that makes Nofa safe: **the AI does the legwork, a person approves anything that spends money or makes a promise.** Below a spend limit some actions can run on their own; everything is logged.

This document lists every task we can automate, which agent owns it, what tools it needs, the guardrails, the data model, the API the website already expects, and a phased plan. Please review it — section 13 lists the decisions I need from you before backend work starts.

> Status: the website (haulix-logistics) is built with a working chat + voice widget in demo mode. The widget calls POST /api/agent/chat, which returns 501 until the backend exists. Nothing in this document is built on the server yet.

02 Who uses it

Person	Their day today	What Haulix takes off their plate
Ops / dispatch coordinator	Retypes emails into the TMS, phones carriers, juggles driver WhatsApps	Order intake, carrier quotes, dispatch, status chasing
Customer service	"Where is my parcel?" calls and emails all day	Tracking answers on chat, voice, WhatsApp, email — with real ETAs
Warehouse lead	Spreadsheets for slots, waves and reorders	Wave plans, slotting, low-stock reorders for approval
Customs / docs clerk	Builds invoices, packing lists, HS codes by hand	Drafts every document for review, flags mismatches
Finance	Matches PODs to invoices, chases disputes	POD matching, invoice drafts, claim packs
Shipper (the customer)	Waits on hold for quotes and updates	Self-serve quotes, bookings, tracking, claims — 24/7
Driver	Paper job sheets, calls the office for changes	Job sheets and route changes pushed to WhatsApp / driver app

03 What can be automated

Autonomy levels: **Auto** = agent acts alone (read-only or low-risk); **Propose** = agent prepares, a person taps approve; **Auto ≤ limit** = auto below a configured spend limit, otherwise Propose; **Human** = agent only

assists.

Order to cash

Task	What the agent does	Agent	Autonomy
Order intake	Reads emails, PDFs, WhatsApp, calls and web forms; extracts shipper, consignee, cargo, dims, dates; asks for what's missing	Intake	Auto (draft)
Address & data validation	Geocodes addresses, checks postcodes, flags dangerous goods keywords, dedupes customers	Intake	Auto
Quoting	Prices the lane across contracted carriers and rate cards; compares price, transit time, CO_2	Quoting	Auto
Booking with carrier	Books the chosen option through carrier API / portal / email	Quoting	Auto ≤ limit
Pickup scheduling	Offers pickup windows, books them, sends labels and job sheet	Dispatcher	Propose
Route planning	Multi-stop routes with time windows, capacity, driver hours; re-plans when orders change	Route planner	Auto (plan), Propose (publish changes after dispatch)
Driver assignment	Matches loads to drivers/trucks by location, hours, licence, vehicle	Dispatcher	Propose
Proof of delivery	Collects signature, photo, GPS; checks counts	Billing & claims	Auto
Invoicing	Drafts invoices from quote + POD + accessorials; checks against the quote	Billing & claims	Propose
Payment chasing	Polite reminders on schedule, answers invoice questions	Billing & claims	Auto

Visibility and exceptions

Task	What the agent does	Agent	Autonomy
Tracking aggregation	Pulls carrier events, AIS (ocean), flight status, GPS; one timeline per shipment	Exception	Auto
Predictive ETA	Estimates arrival from live events, port congestion, weather, history	Exception	Auto
Risk detection	Flags at-risk shipments early (delay, missed connection, customs hold, temperature)	Exception	Auto
Reroute / rebook	Prepares alternative routes or carriers with cost and new ETA	Exception	Propose

Task	What the agent does	Agent	Autonomy
Proactive customer updates	Tells the customer before they ask, on their preferred channel	Customer comms	Auto (templated), Propose (bad news over a threshold)
"Where is my parcel?"	Answers on web chat, voice, phone, WhatsApp, email using the live timeline	Customer comms	Auto
Escalation	Hands hard cases to a person with a summary and suggested reply	Customer comms	Auto

Warehouse

Task	What the agent does	Agent	Autonomy
Inbound / ASN check	Matches arrivals to ASNs and POs, flags shortages and damage	Warehouse	Auto
Slotting	Moves fast movers near the dock, groups items picked together	Warehouse	Propose
Wave / pick planning	Builds pick waves around truck departure times	Warehouse	Auto
Replenishment	Forecasts stock-outs, drafts purchase orders	Warehouse	Propose
Cycle counts	Schedules counts where variance is likely	Warehouse	Auto

Documents and compliance

Task	What the agent does	Agent	Autonomy
Commercial invoice / packing list	Drafts from order data, checks totals and weights	Customs & docs	Propose
HS code suggestion	Suggests codes with reasoning and confidence; never final without a person	Customs & docs	Human (assist)
Bill of lading / AWB / CMR	Pre-fills from booking	Customs & docs	Propose
Document mismatch check	Compares invoice vs packing list vs booking	Customs & docs	Auto
Claims	Assembles photos, timeline, POD, carrier liability; drafts the claim letter	Billing & claims	Propose

04 The agents

One **orchestrator** talks to the person (customer or staff) and routes work to specialist agents. Each specialist owns a small set of tools, so its prompt stays short and its actions are easy to audit. All agents read and write one shared Shipment record — nobody retypes anything.

Agent	Owns	Main tools (functions)
Orchestrator	Conversation, identity, routing, hand-offs	identify_customer, route_to_agent, escalate_to_human
Intake	Turning messages into draft orders	parse_document, extract_order, validate_address, request_missing_info, create_order_draft
Quoting	Rates and carrier bookings	get_rates(lane, cargo), compare_options, create_quote, book_carrier (gated)
Route planner	Routes and re-plans	optimise_routes(stops, fleet, windows), insert_stop, publish_plan (gated)
Dispatcher	Drivers, trucks, job sheets	list_available_drivers, assign_load (gated), send_job_sheet
Warehouse	Waves, slots, stock	get_inventory, forecast_stockout, plan_wave, propose_reorder (gated)
Exception	Risk and recovery	get_tracking_events, predict_eta, assess_risk, propose_reroute (gated)
Customer comms	Updates and answers	get_shipment_timeline, send_update(channel), draft_reply
Customs & docs	Paperwork	draft_commercial_invoice, suggest_hs_code, check_document_set
Billing & claims	Money in, claims out	match_pod, draft_invoice (gated send), build_claim_pack

How a reroute flows (example)

```
carrier event: vessel delayed at Singapore (+14 h)
  -> Exception agent: assess_risk(HX-2041) = high, ETA misses customer deadline
  -> propose_reroute(): options via Colombo (+$120, -9 h) / wait (+$0, +14 h)
  -> Proposal P-311 saved: status=pending, expires in 4 h, needs role=ops_manager
  -> ops manager taps Approve in the ops desk (or replies YES on WhatsApp)
  -> server re-checks: proposal fresh? price unchanged? capacity still there?
  -> book_carrier() with idempotency key -> carrier confirms -> status=confirmed
  -> Customer comms: send_update(customer, WhatsApp+email, new ETA Nov 18)
  -> Audit log: who asked, which agent, data used, who approved, carrier reference
```

05 Guardrails — what stays with people

- **Agents propose, people approve.** Bookings above the limit, reroutes, publishing new routes after dispatch, reorders, invoices and claims are proposals with an expiry. The server re-checks before acting (same pattern as Nofa's confirm-before-booking).
- **Spend limits per company and per role** (e.g. auto-book under $500; manager over $2,000).
- **Facts come from systems, not the model.** ETAs, prices, stock and statuses come from carrier/GPS/WMS data. The model explains; it never invents a number. Plain code checks every figure in

a proposal against the source.

- **Customs decisions are human.** HS codes and declarations are suggestions with confidence; a licensed person signs off.
- **Every action is logged:** request, agent, tool calls, inputs, outputs, approver, external reference. Searchable and exportable for audits.
- **Idempotency and retries:** every external call carries an idempotency key; webhook retries are de-duplicated.
- **Tenant isolation:** each logistics company's data, carriers and rules are separate; a customer can only see their own shipments after verification (shipment ID + email/phone OTP).
- **Kill switch:** an operator can pause any agent or all automation instantly; work falls back to the human queue.

06 Channels

Channel	Who	How (reusing ZOVAK)
Website chat + voice	Shippers, prospects	The Zovak web widget already on the site; text via /api/agent/chat, voice via the GPT-Live browser flow used in Nofa
Phone	Shippers, drivers	Telnyx number → OpenAI SIP → same agent tools (the working Nofa phone setup)
WhatsApp	Shippers, drivers	Meta WhatsApp Cloud API adapter from Nofa; templates for proactive updates
Email inbox	Shippers, carriers	Shared inbox (Gmail/Outlook) read by Intake; replies drafted or sent
EDI / API	Large shippers, carriers	EDI 204/214/210 or JSON webhooks for orders and status
Driver app / WhatsApp	Drivers	Job sheets, route changes, POD capture
Ops desk (staff portal)	Ops team	Live console like the website's demo: shipments, agent log, approvals queue
SMS	Consignees	Delivery windows; Alpha SMS for Bangladesh as planned for Nofa

07 Data model (first version)

Entity	Key fields
Company (tenant)	id, name, spend_limits, carriers[], rate_cards[], settings
Customer / Shipper	id, company_id, name, contacts, addresses, preferred_channel, credit_terms

Entity	Key fields
Order	id, customer_id, source (email/whatsapp/api…), raw_message_ref, status (draft→confirmed), cargo[], pickup, delivery, deadlines
Shipment	id (HX-…), order_id, mode, carrier, legs[], status, eta, eta_confidence, risk_level, documents[]
Leg	shipment_id, from, to, carrier_ref, planned/actual times, vehicle/vessel/flight
TrackingEvent	shipment_id, source, code, location, time, raw
Quote	id, lane, cargo, options[{carrier, price, transit, co2}], chosen, valid_until
Route / Stop	route_id, vehicle_id, driver_id, stops[{order_id, window, eta, seq}], status
Driver / Vehicle	id, licence, hours_used, location, capacity, type
InventoryItem / Bay	sku, bay, qty, min, velocity, supplier
Document	shipment_id, type (CI, PL, BL, AWB, POD, claim), version, status, file_ref
Proposal	id, kind (reroute/booking/reorder/invoice…), payload, cost, needs_role, status, expires_at, approved_by
AuditEvent	time, actor (agent/person), action, inputs, outputs, proposal_id, external_ref
Conversation / Message	channel, customer_id, messages, summary (shared with ZOVAK schema)

08 API the website already expects

The website widget is wired for this contract. Turning on NEXT_PUBLIC_AGENT_LIVE=true and pointing the Next.js route at the backend makes the site live without touching the UI.

```
POST /api/agent/chat            (Next.js route -> proxies to the Haulix backend, like Nofa -> ZOVAK)
request:  { "message": "Any delays on my shipments?", "conversationId": "web-…",
            "history": [{ "role": "user"|"agent", "text": "…" }] }
response: { "text": "…", "quick": ["Reroute HX-2041", "…"],
            "cards": [ Card, … ] }

Card =
 | { kind: "shipment", id, from, to, status, eta, risk?, steps: [{label, done, when?}] }
 | { kind: "quote", lane, options: [{ name, price, days, co2, best? }] }
 | { kind: "proposal", title, detail, confirm, done }      // needs a human tap
```

Backend endpoints (proposed)

Method & path	Purpose
POST /v1/chat (stream)	Conversation turn; streams text, then cards

Method & path	Purpose
POST /v1/proposals/{id}/approve I /reject	Human decision; server re-validates then executes
GET /v1/shipments?status=&risk=	Ops desk list
GET /v1/shipments/{id}/timeline	Events + ETA + risk
POST /v1/orders/intake	Email/WhatsApp/API message in → draft order
POST /v1/quotes	Price a lane
POST /v1/routes/optimise	Plan or re-plan routes
POST /v1/webhooks/{carrierItelnyxImetalemail}	Inbound events (signed, de-duplicated)
GET /v1/audit?shipment=	Audit trail
POST /v1/voice/session	Browser voice session (GPT-Live), same as Nofa

09 Integrations

Need	Options (pick per customer)	First choice for MVP
Carrier rates & booking	Carrier APIs (DHL, FedEx, Maersk), aggregators (Freightos, project44 for visibility), rate-card spreadsheets	Rate cards (CSV) + 1 carrier API
Ocean/air tracking	project44, Terminal49, AIS feeds, airline status	One aggregator
Maps & routing	Google Routes / OSRM / OR-Tools for optimisation	OR-Tools + OSRM (free, self-hosted)
Weather & disruption	OpenWeather, port congestion feeds	OpenWeather
WMS / ERP	Existing WMS via API/CSV, or Haulix's own light inventory	CSV import + own tables
Documents	OCR + LLM extraction (PDF, images)	LLM vision extraction
Messaging	Telnyx (voice/SMS), Meta WhatsApp, Gmail/Outlook, Alpha SMS (BD)	Reuse Nofa adapters
Accounting	Xero, QuickBooks, Tally	Export first

10 What we reuse from Nofa / ZOVAK

ZOVAK piece	Reused for Haulix as
Channel adapters (web chat, voice, phone SIP, WhatsApp, SMS, email)	Same adapters, new tenant config
Shared Python agent + tool-calling loop with streaming	Orchestrator + specialist agents
Read / Propose / Mutate tool policy, proposals with expiry	Bookings, reroutes, reorders, invoices
Plain-code checks after the model (no invented slots/prices)	Price, ETA and stock checks before any card is shown
Outbox worker + notifications	Customer updates, driver job sheets, reminders
Recovery workflow (staff absence → reschedule many bookings)	Exception workflow (one delay → re-plan many shipments)
Staff portal pattern	Ops desk with approvals queue
Audit schema, idempotency, webhook de-duplication	Same, with carrier references
Next.js proxy route + private demo gate	Same for the Haulix site

> New pieces Haulix needs that Nofa doesn't have: a route optimiser, carrier/tracking connectors, document extraction and generation, inventory, and an ETA/risk model.

11 Roadmap

Phase	Scope	Done when
0 · Demo site (done)	Website, 3D globe, scroll story, ops console mock, chat + voice widget in demo mode	Live on a URL you choose
1 · Tracking + customer comms MVP	Shipment/Event tables, CSV + 1 tracking aggregator, "where is my parcel" on web chat, voice and WhatsApp, ops desk read-only	Real shipments answered end-to-end with real ETAs
2 · Intake + quoting	Email/WhatsApp intake to draft orders, rate cards, quote cards, booking proposals with spend limits	An emailed request becomes an approved booking without retyping
3 · Exceptions	Risk scoring, reroute proposals, proactive updates, approvals queue, audit	A delay triggers a proposal and customer update within minutes
4 · Routes + dispatch	OR-Tools routing, driver assignment, job sheets on WhatsApp, POD capture	Daily routes planned and re-planned by the agent
5 · Warehouse, docs, billing	Waves, reorders, CI/PL drafts, HS suggestions, invoice + claim packs	Order-to-cash closed with approvals

12 How we measure it

Metric	Target for a pilot customer
Order intake time (message → booking draft)	< 1 minute
Quote turnaround	< 2 minutes, 24/7
"Where is my parcel" handled without a person	≥ 70%
At-risk shipments flagged before the customer notices	≥ 80%
Km per drop after route optimisation	−10 to −18%
Invoices sent with a POD/quote mismatch	0
Approvals answered within SLA	≥ 95% within 30 min

13 Risks

Risk	Mitigation
Carrier data is late or wrong	Show source and time on every status; confidence on ETAs; human escalation
Model invents a price/ETA	Cards built from database values only; model text is checked against them
Wrong customs code → fines	HS codes are suggestions; licensed person signs off
Too many approval taps	Spend limits and per-action auto rules, tuned per customer
Integration effort per customer	Start with CSV/email + one aggregator; add APIs per pilot
Privacy of shipper data	Tenant isolation, verified identity, retention policy, encrypted secrets

14 Decisions I need from you

- **Target customer for the pilot:** freight forwarder, courier/last-mile, or 3PL warehouse? (Changes which phase comes first.)
- **Region first:** Bangladesh/India (Alpha SMS, local carriers) or international lanes?
- **Separate repo or inside ZOVAK?** My suggestion: a new haulix-backend repo that imports shared ZOVAK packages (channels, agent loop, proposals), so Nofa stays untouched.
- **Spend limits and roles** to start with (e.g. auto ≤ $500, manager ≥ $2,000).
- **Which tracking aggregator / carrier** to integrate first (cost vs coverage).
- **Domain and deploy:** e.g. haulix.zovak.tech on Vercel, like Nofa.